

Multi-Asset Stochastic Pricing & XVA Risk Engine

Documentation: Technical Architecture, Mathematical Foundations,
and Implementation

Study-desk revision — aligned to the code

Udayveer Singh Andotra

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Contents

1	Introduction, Motivation, and Real-World Scenario	3
1.1	The Regulatory and Market Context	3
1.2	Project Motivation and Scenario Description	3
2	System Architecture & Codebase Structure	3
3	Phase 1: Multi-Asset Simulation Engine	4
3.1	Geometric Brownian Motion and Euler–Maruyama	4
3.1.1	Itô’s lemma and log-normal path updating	4
3.2	Stochastic variance (Heston-style CIR layer)	5
3.2.1	Full truncation	5
3.3	Merton jump-diffusion	5
3.4	Cholesky decomposition and spectral repair	5
3.5	Optional Vasicek short rate	5
4	Phase 2: Yield Curve	6
4.1	Instrument tape	6
4.2	Bootstrapping mechanics	6
4.3	NSS versus piecewise interpolation	6
5	Phase 3: Exposure and XVA	6
5.1	Exposure metrics	7
5.2	CVA and FVA	7
5.3	Wrong-way risk and overflow	7
6	Phase 4: Hybrid Portfolio Optimisation	8
6.1	Risk score	8
6.1.1	Why 15 and 8	8
6.2	Zero-exposure trap and growth penalty	8
6.3	Solver	8
7	Interactive Dashboard	8

8	Tests	8
9	Why-nots and remaining extensions	9
9.1	Why not a calibrated Heston surface?	9
9.2	Why not Vasicek as a term-structure model?	9
9.3	Why not full dual-curve OIS / projection?	9
9.4	Why not KVA?	9
9.5	Why not netting, margin, DVA, CSA?	9
9.6	Why not a live market tape?	10
9.7	Why not “enterprise / Basel / client execution”?	10
10	How to read a run	10

1 Introduction, Motivation, and Real-World Scenario

1.1 The Regulatory and Market Context

Following the 2008 global financial crisis, the structural architecture of over-the-counter (OTC) derivatives markets underwent a permanent paradigm shift [4]. Historically, trading desks priced derivative contracts under a risk-free assumption, treating counterparty default risk as a negligible externality. The realization that major financial institutions could default exposed systemic vulnerabilities, leading international regulators to require measurement of Counterparty Credit Risk (CCR) and Valuation Adjustments (XVA) [1].

In that setting, quoting a derivative price without Credit Value Adjustment (CVA) or Funding Value Adjustment (FVA) is incomplete. Banks quantify the expected future cost of counterparty default and of funding uncollateralised exposure across multi-year horizons [4].

This repository is a *study desk* that implements a thin slice of that pipeline on a toy option, with tests for the identities that are actually coded. It is not a production CCR system and it does not implement a Basel capital stack.

1.2 Project Motivation and Scenario Description

Standard textbook models often rely on constant volatility, a single deterministic rate, and a risk-free counterparty. Those assumptions fail under stress. The operational scenario here is narrower than a tier-1 trading book:

A long-only basket of liquid cash equities (for example NSE names) is revalued under Baseline, Bullish, and Market Crash path measures. A European call or put is written on the *portfolio NAV*. The desk reports Expected Exposure (EE), 95% Potential Future Exposure (PFE), unilateral CVA, and a simple uncollateralised FVA, then optionally reweights the book on a demo risk score.

The four phases below match that scenario:

1. Simulate correlated, fat-tailed paths under the three scenarios.
2. Strip a file-driven deposit/swap tape and fit Nelson–Siegel–Svensson (NSS) zeros. Optionally replace deterministic discount factors with Vasicek short-rate paths.
3. Compute EE, 95% PFE, pathwise CVA, and FVA on the toy option.
4. Run a short global-then-local optimiser on a PFE/CVA/FVA score.

2 System Architecture & Codebase Structure

Separation of concerns is modular. “High-performance” here means batched PyTorch tensors on CPU, with CUDA or MPS if present. CuPy is not used.

Module path	Responsibility
<code>core/monte_carlo.py</code>	Correlation Cholesky with spectral repair; CIR variance layer; Merton jumps; optional Vasicek r_t [1, 2, 3].
<code>core/yield_curve.py</code>	Cash-then-swap strip and NSS fit for named tapes <code>discount</code> and <code>projection</code> [4].
<code>data/curve_instruments.csv</code>	Instrument quotes (<code>curve</code> , <code>kind</code> , <code>maturity</code> , <code>rate</code>).
<code>core/xva_calculator.py</code>	Call/put on NAV; EE; 95% PFE; pathwise CVA; FVA; reduced-form WWR.
<code>engine_backend.py</code>	yfinance history, empirical drift/covariance, scenarios, differential evolution and SLSQP [7, 10].
<code>app.py</code>	Streamlit desk: session state, scenario, Vasicek checkbox (off by default), status log [8].
<code>tests/</code>	Identities in §8.

3 Phase 1: Multi-Asset Simulation Engine

The foundation of the risk run is the stochastic core. Before marking the option or integrating CVA, the engine must generate future market states [1].

3.1 Geometric Brownian Motion and Euler–Maruyama

The continuous-time SDE for geometric Brownian motion is

$$dS_t = \mu S_t dt + \sigma S_t dW_t. \quad (1)$$

Computers operate in discrete time. Euler–Maruyama yields

$$S_{t+\Delta t} = S_t + \mu S_t \Delta t + \sigma S_t \sqrt{\Delta t} Z, \quad Z \sim N(0, 1), \quad (2)$$

with $\Delta t = 1/252$ for a daily step [1].

3.1.1 Itô’s lemma and log-normal path updating

Simulating the arithmetic scheme over multi-year horizons introduces a downward discretisation bias. The engine applies Itô’s lemma to $X_t = \ln S_t$:

$$dX_t = \left(\mu - \frac{1}{2} \sigma^2 \right) dt + \sigma dW_t. \quad (3)$$

The implemented update is

$$\ln S_{t+\Delta t} = \ln S_t + \left(\mu - \frac{1}{2} v_t - \lambda k \right) \Delta t + \sqrt{v_t} \Delta W_t^S + J \Delta N_t, \quad (4)$$

then $S_{t+\Delta t} = \exp(\ln S_{t+\Delta t})$. That guarantees positivity. The $\frac{1}{2} v_t$ correction uses the CIR variance of §3.2, not a constant σ^2 .

3.2 Stochastic variance (Heston-style CIR layer)

Constant volatility misses clustering and the leverage effect. Heston introduces a CIR variance [2]:

$$dS_t = \mu S_t dt + \sqrt{v_t} S_t dW_t^S, \quad (5)$$

$$dv_t = \kappa(\theta - v_t) dt + \xi \sqrt{v_t} dW_t^v, \quad (6)$$

with $\mathbb{E}[dW_t^S dW_t^v] = \rho dt$.

What is implemented. $\kappa = 2$, $\xi = 0.3$, $\rho = -0.7$ are hardcoded. θ is the stressed sample variance of each name. The layer is a variance multiplier on unit Browns. It is not a calibrated Heston surface and is not fitted to options. See §9.

3.2.1 Full truncation

A discrete Euler step can send v below zero. The engine uses a full truncation floor [1]:

$$v_{t+\Delta t} = \max\left(v_t + \kappa(\theta - v_t)\Delta t + \xi \sqrt{\max(v_t, 0)} \sqrt{\Delta t} Z^v, 10^{-6}\right). \quad (7)$$

3.3 Merton jump-diffusion

Jumps arrive as a Bernoulli-in- Δt approximation to a Poisson process of intensity λ , with $J \sim N(\mu_J, \sigma_J^2)$. The drift carries the compensator λk ,

$$k = \mathbb{E}[e^J - 1] = \exp(\mu_J + \tfrac{1}{2}\sigma_J^2) - 1, \quad (8)$$

so the objective drift μ is preserved in expectation [1]. Crash scenarios raise λ and use a negative μ_J .

3.4 Cholesky decomposition and spectral repair

Shocks for N names are correlated through the *correlation* matrix R , not through Σ itself. If $D = \text{diag}(\sigma)$ then $R = D^{-1}\Sigma D^{-1}$. A scenario may push off-diagonals toward 1 or toward 0. If the smallest eigenvalue of R is non-positive, the engine adds εI .

Unit Browns are $Z^S = \varepsilon L^\top$ with $LL^\top = R$. Instantaneous variance v is the only diffusion scale. An earlier revision formed $LL^\top = \Sigma$ and then multiplied by $\sqrt{v}$, which double-counted volatility. That is no longer the implementation.

3.5 Optional Vasicek short rate

Interest rates mean-revert. Vasicek is the Ornstein–Uhlenbeck process [3]

$$dr_t = a(b - r_t) dt + \sigma_r dW_t^r. \quad (9)$$

In this revision the process is *optional* and independent of the equity Browns. r_0 and b default to the short end of the discount strip. When `use_vasicek_discount` is true,

$$DF_t^{(m)} = \exp\left(-\sum_{u \leq t} r_u^{(m)} \Delta t\right). \quad (10)$$

The default desk run uses deterministic NSS zeros (§4). The OU paths are a discounting knob, not a calibrated term-structure model. See §9.

4 Phase 2: Yield Curve

Derivatives require a zero curve to present-value cash-flows [4].

4.1 Instrument tape

Quotes live in `data/curve_instruments.csv` with columns `curve`, `kind`, `maturity`, `rate`. There is no live G-sec or SOFR download. Two names may appear under `curve`:

- **discount** — used for DF on this option payoff;
- **projection** — stripped and NSS-fitted so a later *linear* instrument can project index fixings. It does not enter CVA on the NAV option.

That is two independent strips. It is not a post-2008 dual-curve engine (no basis, no OIS versus LIBOR/SOFR machinery). See §9.

4.2 Bootstrapping mechanics

Cash deposits: $Z(T) = 1/(1 + rT)$, $z(T) = -\ln Z(T)/T$.

Par swaps are added in maturity order. Intermediate zeros are linearly interpolated from knots already stored. The last discount factor is the usual par identity

$$Z(T_n) = \frac{1 - S\Delta \sum_{j < n} Z(T_j)}{1 + S\Delta}. \quad (11)$$

In general, instruments with cash-flows C_{ij} and prices P_i satisfy

$$P_i = \sum_j C_{ij} Z(t_j). \quad (12)$$

4.3 NSS versus piecewise interpolation

Linear interpolation of zeros is jagged. Cubic splines with natural boundaries are smoother but can oscillate at the ends (Runge). The engine fits Nelson–Siegel–Svensson by Nelder–Mead [5]:

$$y(t) = \beta_0 + \beta_1 \frac{1 - e^{-t/\tau_1}}{t/\tau_1} + \beta_2 \left(\frac{1 - e^{-t/\tau_1}}{t/\tau_1} - e^{-t/\tau_1} \right) + \beta_3 \left(\frac{1 - e^{-t/\tau_2}}{t/\tau_2} - e^{-t/\tau_2} \right). \quad (13)$$

Default discount factors are $DF(t) = e^{-y(t)t}$ from the discount tape.

5 Phase 3: Exposure and XVA

Phase 3 attaches a counterparty to the simulated book [4].

The simulated spots are first collapsed to a long-only NAV $P_t = \sum_i u_i S_{t,i}$. Strike $K = P_0$ times a scenario multiplier. This is a toy option on NAV, not a netting-set mark-to-market.

5.1 Exposure metrics

Credit exposure is the positive part of the derivative value [4]: $E_t = \max(V_t, 0)$, with $V_t = P_t - K$ for a call and $V_t = K - P_t$ for a put.

Expected Exposure is the cross-sectional mean across M paths,

$$\text{EE}(t_i) = \frac{1}{M} \sum_{m=1}^M E_m(t_i). \quad (14)$$

Potential Future Exposure at 95% is the pathwise quantile

$$\text{PFE}_{95}(t_i) = q_{0.95}(\{E_m(t_i)\}_m). \quad (15)$$

EE sits below PFE on every date (tested).

5.2 CVA and FVA

Textbook CVA with a *deterministic* hazard and no pathwise WWR is

$$\text{CVA}_{\text{EE}} = (1 - R) \sum_i \text{EE}(t_i) DF(t_i) PD(t_i). \quad (16)$$

The code does *not* use that line once intensity depends on exposure. It uses the pathwise sum

$$\text{CVA} = \mathbb{E} \left[\sum_t \text{LGD } E_t DF_t PD_t \right], \quad (17)$$

with $\text{LGD} = 1 - R$ and

$$PD_t = e^{-\Lambda_t - \Delta t} - e^{-\Lambda_t}. \quad (18)$$

When $\alpha = 0$ and DF is deterministic the two formulae agree. When WWR is on, they do not. Tests assert CVA is zero if $R = 1$ and that CVA rises in λ_0 and in α .

Uncollateralised FVA is implemented as

$$\text{FVA} = \mathbb{E} \left[\sum_t s_{\text{fund}} E_t DF_t \Delta t \right]. \quad (19)$$

The textbook display that replaces E_t by $\text{EE}(t)$ is the $\alpha = 0$, deterministic- DF special case. There is no CSA and no DVA.

5.3 Wrong-way risk and overflow

Wrong-way risk is the adverse dependence of exposure and credit quality [4]. The desk uses a reduced-form knob

$$\lambda_t = \lambda_0 \exp(\alpha E_t / P_0), \quad (20)$$

with $\alpha E_t / P_0$ clamped at 20. Absolute INR notionals would overflow `torch.exp` if αE_t were used raw; that is why the code divides by P_0 . This is not a structural model of the counterparty.

6 Phase 4: Hybrid Portfolio Optimisation

Phase 4 reweights the long-only book on a demo score. The weights 15 and 8 are scaling choices, not Basel multipliers.

6.1 Risk score

$$\text{Score}(w) = \max_t \text{PFE}_{95}(t) + 15 \cdot \text{CVA} + 8 \cdot \text{FVA} + \text{GrowthPenalty}. \quad (21)$$

6.1.1 Why 15 and 8

Peak PFE is an undiscounted tail of E_t in currency units. CVA and FVA are present-value integrals against small probabilities and a $\sim 1\%$ spread, so they are one to two orders of magnitude smaller. Without scaling, the optimiser would ignore them.

The factor 15 is a *unit-harmonisation* choice so that CVA has a comparable gradient to peak PFE. The factor 8 does the same for FVA, at lower priority than default-driven CVA. They are not a regulatory capital mapping and should not be cited as one.

6.2 Zero-exposure trap and growth penalty

A call $E_t = \max(P_t - K, 0)$ can be driven to zero by collapsing the book. The engine adds

$$\text{GrowthPenalty} = \max(0, P_0 - \mathbb{E}[P_T]) \cdot 5 \quad (22)$$

outside the crash scenario, so the optimiser cannot buy a zero score by annihilating NAV.

6.3 Solver

SciPy two-stage [7]: differential evolution on $w_i \in [0, 0.25]$, then SLSQP under $\sum_i w_i = 1$. Short budget. Treat the weights as a picture of the score.

7 Interactive Dashboard

app.py is a Streamlit desk [8], not a production control environment.

- Session state for scenario, capital, and last figures.
- `st.status` log from yfinance through NSS and the optimiser.
- Vasicek pathwise discount checkbox, off by default.
- Matplotlib panels: paths, EE/PFE, discount zeros [9].

8 Tests

PYTHONPATH=. python -m pytest -q checks:

- $\text{EE} \leq \text{PFE}$.
- $\text{CVA} = 0$ if $R = 1$; CVA rises in λ_0 and in α .

- FVA = 0 if $s_{\text{fund}} = 0$; FVA rises in the spread.
- Paths start at S_0 and stay positive.
- Deposit-strip identity; NSS on a flat curve.
- Both CSV tapes load; demo projection long end above discount.
- Vasicek starts at r_0 .
- Higher pathwise short rates cut CVA.

9 Why-nots and remaining extensions

The items below were considered. Implementing them as slogans would put this document ahead of the code again — the failure mode of the pre-revision brochure.

9.1 Why not a calibrated Heston surface?

Section 3.2 writes the Heston SDEs because that is the family of the variance layer. Calibration needs an option tape, a loss, and a report of fit quality. None of that is in the repository. The honest name is “CIR variance layer with fixed (κ, ξ, ρ) .”

9.2 Why not Vasicek as a term-structure model?

Section 3.5 writes the OU SDE because that is what `simulate_vasicek_rates` integrates. The paths are not fitted to bonds, do not price swaptions, and are independent of the equity Browns. A term-structure engine would need that calibration. Until then NSS zeros are the default DF and Vasicek is an opt-in stress.

9.3 Why not full dual-curve OIS / projection?

A dual-curve stack separates the curve that projects index fixings from the curve that discounts collateralised cash-flows, plus a basis [4]. This desk has two CSV names and one payoff that never projects a float. Feeding the projection NSS into CVA on an equity option would pretend the second curve is working. The projection strip is stored for a later linear instrument.

9.4 Why not KVA?

KVA is the cost of holding regulatory capital against the uncollateralised book. It needs a capital rule (SA-CCR or IMM), a cost-of-capital rate, and a lifetime integral of expected capital. Inventing $KVA = c \times PFE$ would be a third dashboard number with no identity and no test. KVA stays off the calculator.

9.5 Why not netting, margin, DVA, CSA?

Exposure is $\max(\text{option on NAV}, 0)$ for one counterparty, unilateral, uncollateralised. Netting sets and IM/VM schedules need legal-agreement inputs this repository does not have.

9.6 Why not a live market tape?

Yahoo already fails some days [10]. A live G-sec or SOFR strip adds another vendor and another silent failure. Quotes in a CSV can be checked in review.

9.7 Why not “enterprise / Basel / client execution”?

Those words describe a control environment: model inventory, audit, limits, collateral. This repository is a Streamlit desk and a pytest suite. The claim that can be defended is: the identities we implemented are the identities we test.

10 How to read a run

EE below PFE on the plot. CVA moves with λ_0 , LGD, and α . FVA moves with the funding spread. Crisis correlation fattens PFE faster than EE. Turning Vasicek on should change CVA like a rate stress, not invent a new product. Optimiser weights are a picture of the score.

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